



A control function approach for testing the usefulness of trending variables in forecast models and linear regression

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ABSTRACT

Many predictors employed in forecasting macroeconomic and finance variables display a great deal of persistence. Tests for determining the usefulness of these predictors are typically oversized, overstating their importance. Similarly, hypothesis tests on cointegrating vectors will typically be oversized if there is not an exact unit root. This paper uses a control variable approach where adding stationary covariates with certain properties to the model can result in asymptotic normal inference for prediction regressions and cointegration vector estimates in the presence of possibly non-unit root trending covariates. The properties required for this result are derived and discussed.

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1. Introduction

A common problem in constructing forecasting regressions is understanding the sampling uncertainty of the coefficients in the prediction regression when the predictors appear to be persistent. Forecasters care about this sampling uncertainty because it is often unclear whether or not the predictors are useful for forecasting. Given the near unit root behavior of the predictors, standard asymptotic normal distributions have been shown to be very poor approximations to the sampling distributions of the coefficients in the prediction regression. Such approximations lead to oversized tests and are hence potentially misleading as guides as to whether or not the variables actually have any predictive usefulness.

The classic example in the forecasting literature is the prediction of stock market returns with the dividend price ratio—the dividend price ratio appears to have a root on or near the unit circle. For this regression the problem extends to other popular predictors such as interest rate differentials and the earnings price ratio as well. Similar issues arise with forecasting exchange rates with the forward premium, forecasting changes in income or consumption with interest rate differentials and ratios of macroeconomic variables.

A number of approaches have been taken to provide tests for inclusion of the predictive variables that control size even when the predictive variable has a unit root or near unit root. The common tool has been to use local to unity asymptotics — using

limit theory for sequences of models where the largest root for the predictive regression remains in the neighborhood of one — to approximate distributions. The tests then differ in the precise statistic to be computed and how they handle the unknown root. One approach has been to use Bonferroni or related methods (Cavanagh et al., 1995; Lewellen, 2004; Campbell and Yogo, 2006). Alternatively Moreira and Jansson (2006) condition on a sufficient statistic for the root to remove the dependence in their test. None of these methods dominates each other theoretically or in practice. The precise nature of the problem and these methods are reviewed in Section 2 of this paper.

The methods we have so far have some limitations. First, for each of these methods extending the methods beyond the bivariate regression is extremely challenging, and no results exist in the literature for more than a single predictor in the regression. This is in part because the methods themselves are somewhat cumbersome to apply.

This paper suggests a different approach to obtaining tests which control size. In the method presented here additional covariates are added to the regression. The problems of size distortion and inference are shown to depend on a convolution of the parameters of the trending process and nuisance parameters that describe the relationships between the shocks to the forecasting regression and the shocks to the variables used for forecasting. Judiciously chosen, the covariates have the potential to remove the dependence of the hypothesis tests on the nuisance parameters describing the trend, and so provide inference that is robust to lack of knowledge over the trending behavior of the data. The method is discussed in Section 2 in a simplified bivariate case and examined for very general models in Section 3.

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The method is applicable for a very wide set of problems. First, we allow for the inclusion of a general number of regressors in the prediction regression, hence we are not (as in the methods above) restricted to a single regressor. Second, for the prediction regression the method involves only the running of OLS regressions, and hence even for the general case is straightforward to apply. Third, inference is standard asymptotic (mixed) normal and so hypothesis tests are also straightforward to apply.

We also show, in Section 4, that the method can also be used when examining ‘cointegrating’ regressions where there is uncertainty over a unit root. Such uncertainty is ubiquitous in the cointegration literature, hence the use of unit root and rank pretests. With use of additional stationary covariates, we show similar results as for the prediction equation problem—asymptotically standard (mixed) normal inference can still apply even when the roots are not exactly on the unit circle.

The suggested method, as in the case in instrumental variables, requires finding additional data with particular properties. Such data are referred to as orthogonalizing covariates. In Section 5 we go into more detail regarding these properties. We suggest how such variables can be found in practice. We also show, in Theorem 5, that the use of such covariates that satisfy these properties have at minimum the same power (and generally better power) as would be obtained if we knew the size of the largest root of the predictor variable. Since all of the methods discussed in Section 2 cannot have better power than the case of the root known, this leads directly to the implication that none of the current methods can have better power against any alternative than the method suggested here when we indeed have orthogonalizing covariates.

Monte Carlo results are presented in Section 6 to examine both size and power properties of the suggested method. We also examine the implications of Theorem 5 for the relationship between these methods and the popular Campbell and Yogo (2006) procedure in terms of power.

Proofs of the results are contained in an Appendix.

2. The testing problem and conditioning on stationary covariates

This paper examines two models which have been treated rather differently in the literature but for which common methods apply. The first regression is where a relatively non-trending variable is regressed on lagged trending variables for purposes of prediction. The second is a cointegrating regression where we are uncertain that there are unit roots in the data. In this case the ‘cointegrating’ vector is a trending variable being predicted by contemporaneous trending variables. These problems are similar in the sense that it is not the behavior of the left-hand side variable (which differs here in properties) but the terms on the right-hand side that dominate the theoretical properties of the estimators and tests. In this sense we are able to generate a solution to both problems from a similar idea, as detailed in following sections. The differences lie mostly in the auxiliary assumptions of the model, and the perceived reasonable coefficients on the trending covariates. The remaining difference is that in terms of constructing methods that deal with uncertainty over the trending process, many papers have appeared that examine the prediction regression whereas only a few have examined the cointegrating regression. We focus on the predictive regression in this review section.

The simplest predictive regression test would be to examine if a variable measured today predicts the outcome we are interested in seeing in the next period. For example there is a large literature on forecasting stock returns using the log of the dividend price ratio in the previous period as a predictor. For inference on predictive

ability we require assumptions on both the properties of the predictor variable and on the errors to the regression to justify any inferential procedure on the predictiveness or lack thereof of the regressor. Consider the model

$$(1 - \rho L)(y_{1t} - \varphi_1) = \varepsilon_{1t} \quad t = 2, \dots, T$$

$$y_{2t} = \varphi_2 + \beta y_{1t-1} + \varepsilon_{2t} \quad t = 2, \dots, T \quad (1)$$

where we observe $\{y_t\}_{t=1}^T = \{[y_{1t}, y_{2t}]\}_{t=1}^T$ where both y_{1t} and y_{2t} are univariate, $\varphi_1, \varphi_2, \beta$ and ρ are unknown parameters, and ε_{1t} and ε_{2t} are unknown residual terms. Define the parameter space for ρ as P . Define $E[\varepsilon_{it}\varepsilon'_{jt}] = \Sigma_{ij}$ for $i, j = 1, 2$. The correlation between ε_{1t} and ε_{2t} is $\delta = \Sigma_{12}/(\Sigma_{11}\Sigma_{22})^{1/2}$. In addition, we must initialize the processes, set the initial value of $\varepsilon_{1t} = \xi$ such that $T^{-1/2}\xi \rightarrow^p 0$. Some of these restrictions are made for expository purposes in this review—in the next section y_{2t} can be multivariate, the specification of the deterministic terms is more general, and we also allow serial correlation in $(1 - \rho L)y_{1t}$.

This regression is typically run to test the notion of unpredictability of y_{2t} – the typical null hypothesis derived from economic theory – so we assume that ε_{2t} is uncorrelated. Regressions of this form appear as noted above in the prediction of stock prices by such covariates as the dividend price ratio, earnings price ratio and interest rates. Fama and French (1988) for example reject the hypothesis that the dividend price ratio cannot predict returns. Similar issues arise in testing term structure models of the interest rate, where the lagged interest rate spread is used to predict interest rates. Fama (1975) uses interest rates to predict inflation. Hall (1978) examines the predictability of changes in consumption with lagged income and stock prices. Large literatures reexamining these findings has followed these seminal papers.

In each of these cases, the regressor is typically a persistent or trending variable in the sense that there is clearly a low frequency component to the data. Unit root or near unit root behavior affects the sampling distribution of least squares estimators for β and also hypothesis tests based on this estimate. We can estimate β by OLS, yielding

$$\hat{\beta} - \beta = \frac{\sum_{t=2}^T y_{1t-1}^m \varepsilon_{2t}}{\sum_{t=2}^T (y_{1t-1}^m)^2}$$

where $y_t^m = y_t - (T-1)^{-1} \sum_{t=2}^T y_t$.

Projecting ε_{2t} onto ε_{1t} yields $\varepsilon_{2t} = (\Sigma_{12}/\Sigma_{11})\varepsilon_{1t} + \varepsilon_{2.1t}$ where $E[\varepsilon_{1t}\varepsilon_{2.1t}] = 0$ by construction and the variance of $\varepsilon_{2.1t}$ is $\Sigma_{22} - \Sigma_{12}^2/\Sigma_{11}$. Hence the above expression can be rewritten as

$$\hat{\beta} - \beta = \frac{\sum_{t=2}^T y_{1t-1}^m \varepsilon_{1t} (\Sigma_{12}/\Sigma_{11})}{\sum_{t=2}^T (y_{1t-1}^m)^2} + \frac{\sum_{t=2}^T y_{1t-1}^m \varepsilon_{2.1t}}{\sum_{t=2}^T (y_{1t-1}^m)^2}$$

$$= (\hat{\rho} - \rho)(\Sigma_{12}/\Sigma_{11}) + \frac{\sum_{t=2}^T y_{1t-1}^m \varepsilon_{2.1t}}{\sum_{t=2}^T (y_{1t-1}^m)^2} \quad (2)$$

where $\hat{\rho}$ is the least squares estimator for ρ using data on y_{1t} .

A number of authors have pointed out issues with inference over β in regressions that take this form. Examining the predictability of changes in consumption using lagged income as a predictor, Mankiw and Shapiro (1986) showed in Monte Carlo experiments that hypothesis tests on β did not control size when standard critical values were employed. Stambaugh (1999) showed a similar result for prediction of returns with the dividend

price ratio, using one term expansions of the OLS estimator to attempt to explain the problems with inference.

Since y_{1t-1} and $\varepsilon_{2,t}$ are orthogonal by construction, it is apparent from Eq. (2) that the bias arises from the first term, $(\hat{\rho} - \rho)\Sigma_{12}/\Sigma_{11}$, which is nonzero when both (a) there is persistence, so $\hat{\rho}$ is a biased estimator for ρ , and (b) when there is simultaneity, so $\Sigma_{12} \neq 0$.

Such problems have in many cases been characterized as a ‘small sample’ problem. The idea is that as the sample size increases, $\hat{\rho}$ is consistent for ρ and the problem disappears asymptotically. Such a result requires restricting the parameter space P away from the unit circle, for example in Amihud and Hurvich (2004) bias approximations for $(\hat{\rho} - \rho)$ are based on Edgeworth expansions that require this restriction. Whilst authors differ on how seriously to take the possibility of a unit root in the regressor (the failure to reject a unit root does not of course imply the presence of a unit root), roots close to one are often consistent with the data. So when the uncertainty includes a unit root, that there will always be some region of the parameter space of ρ near one for which problems of size control occur regardless of the sample size.

To account for the possibility of unit roots and near unit roots, we model the region near one using the local to unity methods of Phillips (1987) where¹ $\rho = 1 - \gamma/T$. These methods are known to provide much better asymptotic approximations for sample size and ρ pairs such that γ is in the range of 40 to a small positive number (including the unit root process smoothly with models near the unit root). In this approach we define the Brownian motion $W_1(s)$ as the suitably scaled limit of the partial sum of the shocks to y_{1t} and $\bar{M}(s)$ is a demeaned Ornstein–Uhlenbeck process driven by $W_1(s)$ (these are defined more precisely in Section 3). Under such asymptotics it is well known that $T(\hat{\rho} - \rho)$ converges to a nonstandard distribution (a function of $\bar{M}(s)$). Hence the bias remains asymptotically. Indeed, as shown in Elliott and Stock (1994) for this model

$$t_{\hat{\beta}=\beta_0} \Rightarrow \delta \hat{\tau}_{DF} + (1 - \delta^2)^{1/2} \phi \quad (3)$$

where $\hat{\tau}_{DF}$ has a nonstandard asymptotic distribution and ϕ is a standard (mixed) normal random variable. When $\rho = 1$ ($\gamma = 0$), $\hat{\tau}_{DF}$ is the usual Dickey and Fuller (1979) distribution for the t statistic testing for a unit root in y_{1t} . For $\gamma > 0$ this distribution remains nonstandard with similar features to the distribution when $\rho = 1$.

This distributional approximation explains clearly the Monte Carlo findings and suggests problems with standard inference in predictive regressions of this sort. First, as is well known, when $\rho = 1$ the distribution for $\hat{\tau}_{DF}$ has a long left tail. Hence when $\delta < 0$ this corresponds to size distortions using standard critical values in the right tail of $t_{\hat{\beta}=\beta_0}$ —when β is truly zero and the regressor has no predictive power the regression is likely to find $\hat{\beta} > 0$ and statistically significant with a frequency greater than size. The larger in absolute value is the correlation the more likely we are to reject.

The effects can be seen in Fig. 1. The figure shows the upper 95th percentiles of the distribution in (3) for $\delta = -0.95, -0.75$ and -0.5 respectively as a function of γ . These percentiles would be the correct critical values for a one sided 5% test for $\beta = 0$. The problem of rejecting for t statistics greater than 1.645 for a 5% test—those used when using a normal approximation—can be seen from the figure. When ρ is sufficiently close to 1 and the sample size such that $\gamma = T(\rho - 1)$ is sufficiently close to zero then for δ of the sizes often seen in application t statistics of greater than

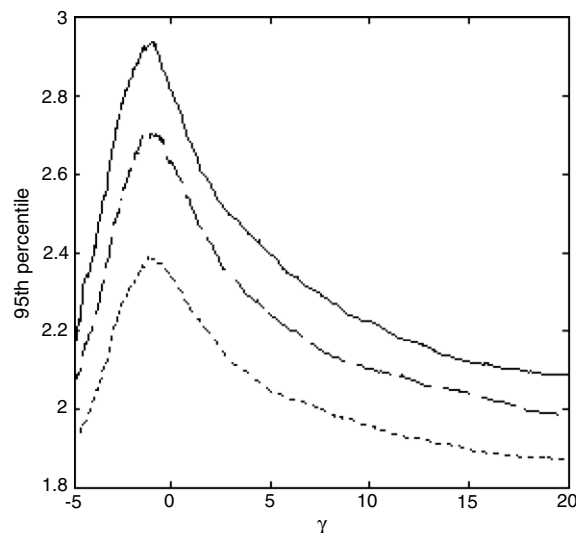


Fig. 1. Curves show 95th percentile of (3) as a function of γ for $\delta = -0.95$ (solid line), $\delta = -0.75$ (long dashes) and $\delta = -0.5$ (short dashes).

2 and even near 3 that would lead one to reject the hypothesis of no predictability are actually consistent with the null hypothesis of no predictability. This translates to tests that are greatly oversized. The distortion is smaller the larger is γ (so y_{1t} is less persistent), where the first term in (3) becomes closer to a standard normal distribution (Phillips, 1987). For smaller values for δ the extent of the oversize of the test statistic becomes smaller, when δ is close to zero there is little problem, with critical values close to their usual values.

Fortunately, since δ is consistently estimable even without knowledge of ρ , in applications we are able to gauge the extent of the problem. The consistency of standard estimates for δ follows directly from Σ being consistently estimable in regression since the parameters converge at a fast enough rate so that estimation error is negligible asymptotically (this is of course why we are able to estimate the variance covariance matrix to normalize t -statistics in typical regression problems). Many of the methods discussed below are derived for known δ and work in practice by replacing this value with a consistent estimate. The discussion for the remainder of this section will assume this parameter is known.

A number of approaches have been suggested to overcome this size distortion problem. An elegant approach is to condition on the sufficient statistic for ρ and construct a test for the null hypothesis based on the conditional likelihood. This approach is detailed in Moreira and Jansson (2006), who provide tests for the bivariate model case. The method involves a numerical integration technique to construct p -values for the test. A limitation of this approach is that it becomes very difficult to extend the method beyond the bivariate model. Stambaugh (1999) suggested a Bayesian approach, which avoids the requirement that the root be defined to a single number. Bootstrap approaches that involve estimating ρ in order to regenerate bootstrap samples of the data $\{y_t^*\}$, though common in the applied literature, result in bootstrap distributions that differ from the sampling distribution for the statistics of interest (see Basawa et al., 1991) and hence do not control size asymptotically and thus do not solve the problem.

An alternative approach that does control size reasonably well, and as such have been more often employed in the applied literature, is to examine the distribution of (3) for all ‘reasonable’ values for ρ and use the most conservative ones. Cavanagh et al. (1995) suggested such a test based on Bonferroni and Scheffe methods which incorporate sample information on likely values for ρ . Such a procedure results in a test that controls size but

¹ In the asymptotic theory we employ triangular asymptotics, so ρ is ρ_T , we follow the econometric literature and ignore the subscript to enhance readability.

results in biased tests. Essentially the critical values may be too conservative for many values of ρ and hence for a particular (ρ, δ) pair numerical size is much less than nominal size. This results in a biased test since there are local alternatives for which the test rejects less often than nominal size. Power is also affected, although these methods have reasonable power for many alternatives.

We can rotate the system so that the model is

$$(1 - \rho L)(y_{1t} - \varphi_1) = \varepsilon_{1t} \\ y_{2t} = \varphi_2 + \beta y_{1t-1} + \lambda(1 - \rho L)y_{1t} + \varepsilon_{2,1t}$$

where $\lambda = \Sigma_{12}/\Sigma_{11}$ and $\varepsilon_{2,1t}$ is by construction orthogonal to ε_{1t} . In this regression the orthogonality of the residuals means that the history of y_{1t} is uncorrelated with the regression residuals. The infeasible SUR problem thus rotates the model so that the regression to be run is

$$y_{2t} = \varphi_2 + \beta y_{1t-1} + \lambda(1 - \rho L)y_{1t-1} + \varepsilon_{2,1t}. \quad (4)$$

In this regression we have that $E[\varepsilon_{2,1t}|y_{1t}] = 0$ and hence the bias has been removed. The t statistic testing hypotheses on β now have an asymptotic mixed normal distribution regardless of ρ . Thus standard critical values from the normal distribution apply asymptotically.

This is of course infeasible since constructing the additional regressor requires that we know ρ in order to construct the quasi-differenced regressor $(1 - \rho L)y_{1t-1}$. Campbell and Yogo (2006) utilize this regression with ρ known to obtain confidence intervals on $\hat{\beta}$ given ρ . They then use a Bonferroni procedure which employs a DF-GLS test for a unit root (Elliott et al., 1996) and the largest bounds from each of the values for ρ in the confidence interval on ρ to construct confidence intervals for $\hat{\beta}$ robust to the lack of knowledge over ρ .

Plugging in an estimator $\hat{\rho}$ to construct the quasi difference results in a correctly centered t -statistic on β having the asymptotic distribution

$$t_{\hat{\beta}_{OLS}=\beta_0} \Rightarrow \frac{\delta}{(1 - \delta^2)^{1/2}} \hat{t}_{DF} + \phi \quad (5)$$

which is again nonstandard when there is simultaneity.

The infeasible SUR result, which results in standard inference on β , does give some insight into the problem. Standard inference would result if we could include the orthogonalizing variable $(1 - \rho L)y_{1t}$ for the regression, however estimates of ρ are not sufficiently precise, even asymptotically. Similar results will obtain for any estimator for ρ that converges at rate T .

However this possibility of obtaining standard inference could still be useful if we had a proxy for the orthogonalizing variable $(1 - \rho L)y_{1t}$ for the regression. In this paper we take a different approach to the problem, one more aligned with solutions that arise from the economics behind the problem. Consider instead a stationary variable z_t that is contemporaneously correlated with both y_{2t} and $(1 - \rho L)y_{2t}$. We could run the regression

$$y_{2t} = \varphi_2 + \beta y_{1t-1} + \alpha_2 z_t + \tilde{\varepsilon}_{2t}$$

where $\tilde{\varepsilon}_{2t}$ is the remainder of a projection of ε_{2t} on z_t . Since the correlation between z_t and y_{2t} is contemporaneous then α_2 nonzero does not contradict theories that imply lack of predictability. Typically economic theory suggests many possible factors that are likely to be contemporaneously related to variables we are interested in predicting. Although it may be difficult to find predictors of stock market returns, explanations abound of news that contemporaneously affects market prices.

The addition of stationary covariates to the model affects the limit distribution for the OLS estimator for β and its attendant t statistic. The distribution for the t statistic in this case is again

a linear combination of a nonstandard term and a mixed normal term, namely

$$t_{\hat{\beta}=\beta_0} \Rightarrow \tilde{\delta} \hat{t}_{DF} + (1 - \tilde{\delta}^2)^{1/2} \phi \quad (6)$$

where $\tilde{\delta}$ is the correlation between $\tilde{\varepsilon}_{1t}$ and $\tilde{\varepsilon}_{2t}$ where $\tilde{\varepsilon}_{it}$, $i = 1, 2$ is the remainder of a projection of ε_{it} on z_t . Hence the inclusion of z_t in the regression changes the weights on the standard and nonstandard pieces.

Thus for any z_t this simply results in yet another mixture distribution which in aggregate is nonstandard. For $|\tilde{\delta}| < |\delta|$ the effect of the nonstandard piece and hence the possibility of size distortions from using the normal are smaller. However it may be possible to find variables z_t — the orthogonalizing covariates — for which $\tilde{\delta} = 0$ even though δ does not. In this case the weight on the nonstandard part of the distribution is zero and the t statistic has a mixed normal distribution with mean zero and variance unity. Hence standard inference with the usual critical values from the normal table is appropriate asymptotically. In Section 3 these points will be made precise in the general model.

Similar problems in inference arise in the cointegrating model when variables do not have an exact unit root. Elliott (1998) examines this problem for estimators of the cointegrating vectors that are efficient when the roots are exactly on the unit circle. The sizes of the distortions can be even larger for the near cointegrating case than that for the predictive case. However a correct orthogonalization can remove the effect. This result is shown for general models in Section 4.

3. Standard inference for predictive regressions with near unit roots

The general predictive model is

$$(1 - \rho L)(y_{1t} - \varphi_1 d_t) = u_{1t} \quad (7)$$

$$y_{2t} = \varphi_2 d_t + \beta y_{1t-1} + \varepsilon_{2t}$$

$$z_t = \varphi_3 d_t + u_{3t}$$

where the observed data $\{y_{1t}, y_{2t}, z_t\}$ are such that y_{1t} is $k \times 1$, y_{2t} is univariate and z_t is $p \times 1$, d_t is a vector of deterministic terms, $A(L)u_{1t} = \varepsilon_{1t}$ and $B(L)u_{3t} = \varepsilon_{3t}$ where $B(L)$ has order q , $\varepsilon_t = (\varepsilon'_{1t}, \varepsilon'_{2t}, \varepsilon'_{3t})'$ where $E[\varepsilon_t \varepsilon'_t] = \Sigma$. It will be useful to partition Σ into blocks Σ_{ij} , $i, j = 1, 2, 3$ which correspond to each of the three elements of ε_t (so $E[\varepsilon_i \varepsilon'_j] = \Sigma_{ij}$). The unknown parameters are $(\varphi_1, \varphi_2, \varphi_3, \beta, \rho, \xi, \Sigma)$ and the parameters of the finite lag polynomials $A(L)$ and $B(L)$.

The model is intended to capture the problem of forecasting a univariate outcome of interest y_{2t} with a $k \times 1$ set of predictive regressors y_{1t-1} . These predictive regressors are trending regressors in the sense that the autoregressive coefficient that corresponds to their largest roots, here factored out into ρ , are close to one in the statistical sense of these roots being difficult to distinguish from either on the unit circle or just outside. We restrict ρ to be a $k \times k$ matrix which is diagonal² with diagonal elements ρ_i for $i = 1, \dots, k$.

In order to obtain asymptotic approximations for sampling distributions of the estimators and tests considered here, we make the following additional assumptions on the model.

Condition 1. (a) The data $\{y_{1t}, y_{2t}, z_t\}$ are a triangular array and $\rho = 1 - T^{-1}\Gamma$ where Γ is a diagonal matrix with diagonal elements γ_i , $i = 1, \dots, k$.

² This restriction ensures that the y_{1t} variables do not have the property that their second differences satisfy a Functional Central Limit Theorem (FCLT) with a positive but finite scale term, i.e. they are not $I(2)$ or near $I(2)$ random variables.

- (b) $E[\varepsilon_t | \varepsilon_{t-1}, \varepsilon_{t-2}, \dots] = 0$, $E[\varepsilon_t \varepsilon_t' | \varepsilon_{t-1}, \varepsilon_{t-2}, \dots] = \Sigma$ and $E[\varepsilon_t^m | \varepsilon_{t-1}, \varepsilon_{t-2}, \dots] < K < \infty$, $m > 4$ for $i = 1, \dots, k+1+p$.
- (c) $A(1)^{-1} \Sigma_{11} A(1)^{-1'}$ has rank k .
- (d) $A(L)$ and $B(L)$ have all roots outside the unit circle.
- (e) $T^{-1/2} \xi \rightarrow^p 0$.
- (f) d_t (which is a function of d_t) is such that there exists a diagonal scaling matrix $\Upsilon_{d,T}$ which is only a function of T such that $\Upsilon_{d,T}^{-1} \tilde{d}_t = T^{-1/2} \tilde{d}_t$ and $T^{-1} \sum_{t=1}^T \tilde{d}_t \tilde{d}_t' \rightarrow \int_0^1 s(\lambda) s(\lambda)' d\lambda$ where $s(\lambda)$ is nonstochastic.

In Condition 1(a) we make formal the use of the local to unity asymptotic theory, following Phillips (1987). This is standard in this literature. Condition 1(b) ensures that the shocks driving the system are regular enough to allow asymptotic inference, in particular $T^{-1/2} \sum_{t=1}^T \varepsilon_{t,T} \Rightarrow \Sigma^{1/2} W(\cdot)$ where $W(\lambda)$ is a $(k+1+p) \times 1$ standard Brownian Motion with variance covariance matrix λI_{k+1+p} . The requirement in Condition 1(c) is that there do not exist any linear combinations of y_{1t} such that their partial sums would satisfy a functional central limit theorem. In the case where $\rho = I$ this condition amounts to assuming that none of the elements of y_{1t} are cointegrated with each other. The condition in (d) is required to restrict asymptotic inference on the coefficients of the z_t covariates to be asymptotic normal converging at rate $T^{1/2}$. This condition of course is the standard condition for asymptotic normal inference when dependence is linear. The initial condition for y_{1t} is made negligible in (e), which is the standard assumption made in this literature. In particular the entire set of results for the predictive regression discussed above make this assumption or the more restrictive one that $\xi = 0$. Condition 1(f) limits the types of deterministic terms that are handled by the results below. The condition allows for a constant, polynomials in time, and many other types of deterministic terms that do not depend on estimated quantities. The key assumption is that $s(\lambda)$ does not depend on estimated quantities but is instead known.³

Condition 1(a), (b), (d), (e), the first equation in (7) along with choosing $\Sigma^{1/2}$ to be lower block diagonal with partitions after the k th row and column yields the asymptotic approximation $T^{-1/2} u_{1[T\lambda]} \Rightarrow A(1)^{-1} \Sigma_{11}^{1/2} M(\lambda)$ where $M(\lambda)$ is a $k \times 1$ standard Ornstein–Uhlenbeck process which is a continuous function of $W_1(\lambda)$ where $W_1(\lambda)$ are the first k elements of $W(\lambda)$. The choice of $\Sigma^{1/2}$ lower block diagonal ensures that $M(\lambda)$ is independent of the remaining elements of $W(\lambda)$.

The standard approach to examining this problem, since it is a problem of examining the predictability of y_{1t} is to estimate directly the prediction equation without regard to the rest of the system. The prediction regression is

$$y_{2t} = \varphi_2' d_t + \beta' y_{1t-1} + \varepsilon_{2t}. \quad (8)$$

The null hypothesis of interest is $H_0 : \beta = 0$ with a two sided alternative. This can be tested using the usual Wald test.

Theorem 1. When Condition 1 holds then under the null hypothesis testing $\beta = 0$ in the regression (8) the Wald test has the limiting distribution

$$\text{Wald} \Rightarrow \left(\int \bar{M} dW_1 \delta + (1 - \delta' \delta)^{1/2} \int \bar{M} dW_2 \right)' \left(\int \bar{M} \bar{M}' \right)^{-1} \times \left(\int \bar{M} dW_1 \delta + (1 - \delta' \delta)^{1/2} \int \bar{M} dW_2 \right)$$

where $\delta = \Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1/2}$ and $\bar{M}(\lambda)$ is the remainder of a projection of $M(\lambda)$ on $s(\lambda)$.

³ An implicit assumption is that the form of $s(\lambda)$ is the same for both (8) and (9). The results are easily amended at the cost of additional notation when these differ.

This result extends that of Elliott and Stock (1994) to the case of more than a single regressor and also allows for a greater variety of deterministic terms. The basic point however is the same—when there is correlation between ε_{1t} and ε_{2t} so Σ_{12} is nonzero then the distribution of any test for the null hypothesis does not have the standard distribution and as a consequence there is typically a size distortion if standard critical values are used. Rejections of the null hypothesis of no predictability may in fact be due to this correlation.

The regressors z_t are available stationary covariates that may be correlated with the shocks to the prediction equation (if Σ_{23} is nonzero) and may be contemporaneously correlated with the shocks to the predictor variables (if the relevant element of Σ_{13} is nonzero).

The regression to be run when the z_t are included in the regression is

$$y_{2t} = \tilde{\varphi}_2' \tilde{d}_t + \beta' y_{1t-1} + \alpha_2' Z_t + \tilde{\varepsilon}_{2t} \quad (9)$$

where $Z_t = (z_t', z_{t-1}', \dots, z_{t-q}')'$, $\alpha_2 = (\alpha_{20}', \alpha_{21}', \dots, \alpha_{2q}')'$ and we define $\tilde{\varepsilon}_t = (\tilde{\varepsilon}_{1t}', \tilde{\varepsilon}_{2t}')'$ as the remainder when we project $(\varepsilon_{1t}', \varepsilon_{2t}')'$ onto ε_{3t} . For partitions conformable with (y_{1t}', y_{2t}', z_t') define the matrix

$$G = \begin{pmatrix} I_k & 0 & -\Sigma_{13} \Sigma_{33}^{-1} \\ 0 & 1 & -\Sigma_{23} \Sigma_{33}^{-1} \end{pmatrix}$$

then $\tilde{\varepsilon}_t = G \varepsilon_t$ and so the population values for α_{2t}' are $\Sigma_{23} \Sigma_{33}^{-1} B_i$ where B_i is the i th element of $B(L)$.

We consider the Wald statistic testing the same hypothesis as in Theorem 1.

Theorem 2. When Condition 1 holds then under the null hypothesis testing $\beta = 0$ in the regression (9)

$$\text{Wald} \Rightarrow \left(\int \bar{M} dW_1 \tilde{\delta} + (1 - \tilde{\delta}' \tilde{\delta})^{1/2} \int \bar{M} d\tilde{W}_2 \right)' \left(\int \bar{M} \bar{M}' \right)^{-1} \times \left(\int \bar{M} dW_1 \tilde{\delta} + (1 - \tilde{\delta}' \tilde{\delta})^{1/2} \int \bar{M} d\tilde{W}_2 \right)$$

where $\tilde{\delta} = \tilde{\Sigma}_{11}^{-1/2} \tilde{\Sigma}_{12} \tilde{\Sigma}_{22}^{-1/2}$, $\tilde{\Sigma} = E[\tilde{\varepsilon}_t \tilde{\varepsilon}_t']$ defined below and $\tilde{W}(\lambda)$ is a standardized linear combination of $W_2(\lambda)$ and $W_3(\lambda)$.

This theorem extends the results to the case where stationary variables are included in the regression. Since $\tilde{W}(\lambda)$ is a linear function of $W_2(\lambda)$ and $W_3(\lambda)$ then $\tilde{W}(\lambda)$ is independent of $\bar{M}(\lambda)$ then $\int \bar{M} d\tilde{W}_2$ is mixed normal. This means that, just as in the previous result, the distributions are a weighted average between a term that is mixed normal and a nonstandard term. The difference lies in the weighting—here the weighting on the nonstandard component is $\tilde{\delta}$ in place of δ . The different weighting means that the extent to which the distribution differs from a standard one will differ.

We can consider the impact of adding the additional covariates on the nuisance parameters though since

$$\begin{pmatrix} \tilde{\Sigma}_{11} & \tilde{\Sigma}_{12} \\ \tilde{\Sigma}_{12}' & \tilde{\Sigma}_{22} \end{pmatrix} = \begin{pmatrix} \Sigma_{11} - \Sigma_{13} \Sigma_{33}^{-1} \Sigma_{13}' & \Sigma_{12} - \Sigma_{13} \Sigma_{33}^{-1} \Sigma_{23}' \\ \Sigma_{12}' - \Sigma_{23} \Sigma_{33}^{-1} \Sigma_{13}' & \Sigma_{22} - \Sigma_{23} \Sigma_{33}^{-1} \Sigma_{23}' \end{pmatrix}. \quad (10)$$

The inclusion of the extra stationary variables can result in $|\tilde{\delta}| > |\delta|$, i.e. even larger size distortions that when z_t is omitted, although it should be recalled that both $\tilde{\delta}$ and δ are consistently estimable so this situation can easily be avoided in practice.

But it is also possible that by including z_t with the correct properties that $\tilde{\delta} = 0$ and standard inference on the OLS coefficients provides correct inference asymptotically. What is required is that the control variables z_t be ‘orthogonalizing covariates’ and explain away the correlation between ε_{1t} and ε_{2t} . Mathematically

this means that $\Sigma_{12} = \Sigma'_{13}\Sigma_{33}\Sigma_{23}$. This makes clear two necessary properties that must be satisfied by z_t —they are (a) they must be stochastic and (b) both Σ_{13} and Σ_{23} must be nonzero. The second of these means that z_t must covary with both the dependent variable and the quasi difference of the predictor variables. Even when $\tilde{\delta}$ is nonzero it may be close enough to zero so that size distortions are small.

4. Standard inference for cointegrating relationships with near unit roots

The general near cointegrating model is

$$\begin{aligned}(1 - \rho L)(y_{1t} - \varphi_1 d_t) &= u_{1t} \\ y_{2t} &= \varphi_2 d_t + \beta y_{1t} + u_{2t} \\ z_t &= \varphi_3 d_t + u_{3t}\end{aligned}\quad (11)$$

where $u_t = (u'_{1t}, u'_{2t}, u'_{3t})'$ and $\tilde{A}(L)u_t = \varepsilon_t$. The observed data $\{y_{1t}, y_{2t}, z_t\}$ are such that y_{1t} is $k \times 1$, y_{2t} is $r \times 1$ (so there are r potential cointegrating vectors) and z_t is $p \times 1$, d_t is a vector of deterministic terms, $\varepsilon_t = (\varepsilon'_{1t}, \varepsilon'_{2t}, \varepsilon'_{3t})'$ where $E[\varepsilon_t \varepsilon'_t] = \Sigma$, and the unknown parameters are $(\varphi_1, \varphi_2, \varphi_3, \beta, \rho, \xi, \Sigma)$ and the parameters of the finite (known) lag polynomial $\tilde{A}(L)$.

This model can be rotated into the vector autoregressive model

$$\begin{pmatrix} \Delta y_{1t} \\ \Delta y_{2t} \\ z_t \end{pmatrix} = \Psi \begin{pmatrix} y_{1t-1} \\ y_{2t-1} \\ z_{t-1} \end{pmatrix} + \Phi \bar{d}_t + \Pi(L) \begin{pmatrix} \Delta y_{1t-1} \\ \Delta y_{2t-1} \\ z_{t-1} \end{pmatrix} + \varepsilon_t^* \quad (12)$$

where Ψ is $(k+r+p)x(k+r)$ and $\bar{d}_t$ are the relevant deterministic terms that derive from (11). For deterministic terms that are a polynomial of time this is straightforward as $\bar{d}_t = d_t$, however if there are known structural breaks then this term can become more complicated. It is assumed throughout that the deterministic component is correctly specified in the sense that no terms implied by the original model have been omitted.

In the case where $\rho = I_k$ and we assume that the elements of y_{1t} themselves do not cointegrate amongst each other, this is a general cointegrating system where if we partition $\Psi = [\Psi_1 \Psi_2]$ where the partition is after the k th column then Ψ_2 are the impact coefficients of the error correction terms $(y_{2t-1} - \beta y_{1t-1})$ where β is the cointegrating vector. When the z_t covariates are omitted a wide range of asymptotically equivalent methods are available under this assumption for estimation of and inference on the cointegrating vector (cf. Johansen, 1988, 1991; Stock and Watson, 1993; Saikkonen, 1991, 1992; Phillips and Hansen, 1990). Hypothesis tests on the cointegrating vector using these methods is particularly simple since t -statistics can be compared to the usual normal distribution and more general tests can be compared to the χ^2 distributions.

Relaxing the condition that we know a priori that the roots are exactly on the unit circle (and maintaining the assumption that we ignore the z_t covariates) Elliott (1998) shows that the efficient methods for estimating the cointegrating vector have a bias term and standard inference no longer applies except in the very special case where the correlation between u_{1t} and u_{2t} is zero at the zero frequency.

For estimating $\hat{\beta}$ in this regression we consider the estimator

$$\hat{\beta} = -(\hat{\psi}'_2 \hat{\Sigma}^{*-1} \hat{\psi}_2)^{-1} (\hat{\psi}'_2 \hat{\Sigma}^{*-1} \hat{\psi}_1) \quad (13)$$

where the rows of $\hat{\psi} = [\hat{\psi}_1 \hat{\psi}_2]$ are obtained from running each of the regressions in (12) equation by equation (as in the usual case when we estimate a VAR) and $\hat{\Sigma}^* = T^{-1} \sum \hat{\varepsilon}_t^* \hat{\varepsilon}_t^{*'} where $\hat{\varepsilon}_t^*$ are the least squares residuals from the OLS regressions (12).$

This estimator is an extension of the method proposed by Saikkonen (1992) for the case where there are no stationary

variables. The Granger representation theorem shows that when $\rho = I_k$ then cointegration implies that the matrix Ψ is reduced rank. This estimator can be shown to be asymptotically equivalent to the minimum distance estimator for the cointegrating vector when the restrictions implied by cointegration are imposed on the unrestricted VAR estimates. Further, following Saikkonen (1992) this estimator can be shown to be asymptotically equivalent to a maximum likelihood estimator for the same normalization of the cointegrating vector.

We turn to the asymptotic properties of this estimator. Additional assumptions on the model are essentially the same as Condition 1, apart from (c) and (d).

- Condition 2.** (a) The data $\{y_{1t}, y_{2t}, z_t\}$ are a triangular array and $\rho = 1 - T^{-1}\Gamma$ where Γ is a diagonal matrix with diagonal elements γ_i , $i = 1, \dots, k$.
 (b) $E[\varepsilon_t | \varepsilon_{t-1}, \varepsilon_{t-2}, \dots] = 0$, $E[\varepsilon_t \varepsilon'_t | \varepsilon_{t-1}, \varepsilon_{t-2}, \dots] = \Sigma$ and $E[\varepsilon_{it}^m | \varepsilon_{t-1}, \varepsilon_{t-2}, \dots] < K < \infty$, $m > 4$ for $i = 1, \dots, k+1+p$.
 (c) $\tilde{A}_{11}(1)^{-1} \Sigma_{11} \tilde{A}_{11}(1)^{-1'}$ has rank k where $\tilde{A}_{11}(L)$ is the upper $k \times k$ block of $\tilde{A}(L)$.
 (d) $\tilde{A}(L)$ has all roots outside the unit circle.
 (e) $T^{-1/2} \xi \rightarrow^p 0$.
 (f) $\bar{d}_t$ (which is a function of d_t) is such that there exists a diagonal scaling matrix $\Upsilon_{d,T}$ which is only a function of T such that $\Upsilon_{d,T}^{-1} \bar{d}_t = T^{-1/2} \tilde{d}_t$ and $T^{-1} \sum_{t=1}^T \tilde{d}_t \tilde{d}_t' \rightarrow \int_0^1 s(\lambda) s(\lambda)' d\lambda$ where $s(\lambda)$ is nonstochastic.

Theorem 3. For the model in (11) when $\rho = I_k - T^{-1}\Gamma$ and Condition 2 holds then for the estimator defined in (13) then

$$\begin{aligned}T(\hat{\beta} - \beta) &\Rightarrow \tilde{\Omega}_{2,1} \iota'_{2'} \Omega^{-1/2'} \int dW \bar{M}' \left(\int \bar{M} \bar{M}' \right)^{-1} \\ &\quad \times \Omega_{11}^{-1/2} + \tilde{\Omega}_{12}' \tilde{\Omega}_{11}^{-1} \Gamma\end{aligned}\quad (14)$$

where $\Omega = A(1)^{-1} \Sigma A(1)^{-1'}$, $\tilde{\Omega}_{ij} = \Omega_{ij} - \Omega_{i3} \Omega_{33}^{-1} \Omega_{3j}$, $\tilde{\Omega}_{2,1} = \tilde{\Omega}_{22} - \tilde{\Omega}_{12}' \tilde{\Omega}_{11}^{-1} \tilde{\Omega}_{12}$, $\bar{M}$ is as defined in the text after Condition 1.

This result extends the results of Saikkonen (1991) to the case where stationary covariates are included in the regression and used optimally to estimate the cointegrating vector under the assumption that $\rho = I_k$. Since the leading $r \times k$ block of $\iota'_{2'} \Omega^{-1/2'}$ is a matrix of zeros this means that in the weighted average of $\int dW \bar{M}'$ a zero weight is given to $\int dW_1 \bar{M}'$ (which is a nonstandard piece) and the remaining pieces are mixed normal. Hence the first term in (14) has a mixed normal distribution.

Linear hypotheses on β in the form $R\text{vec}(\beta) = r$ can be tested using the Wald test

$$\begin{aligned}\text{Wald} &= T^2 (R\text{vec}(\hat{\beta}) - r)' \left[R(\hat{M} \otimes (\hat{\psi}'_2 \hat{\Sigma}^{*-1} \hat{\psi}_2)^{-1})^{-1} R' \right]^{-1} \\ &\quad \times (R\text{vec}(\hat{\beta}) - r)\end{aligned}$$

where $\hat{M}$ is the upper $k \times k$ block of the inverse of the regressors outer product scaled by T^{-2} . This is the standard form of the Wald statistic apart from the scaling term $(\hat{\psi}'_2 \hat{\Sigma}^{*-1} \hat{\psi}_2)^{-1}$ in place of the variance covariance of the residuals.

Theorem 4. Under the same conditions as Theorem 3 then under the null hypothesis that β is true

$$\begin{aligned}\text{Wald} &\Rightarrow \left\{ R\text{vec} \left(\tilde{\Omega}_{2,1}^{1/2'} \int dW \bar{M}' \left(\int \bar{M} \bar{M}' \right)^{-1} \Omega_{11}^{-1/2} + \tilde{\Omega}_{12}' \tilde{\Omega}_{11}^{-1} \Gamma \right) \right\}' \\ &\quad \left[R \left(\Omega_{11}^{-1/2'} \left(\int \bar{M} \bar{M}' \right)^{-1} \Omega_{11}^{-1/2} \otimes \tilde{\Omega}_{2,1} \right) R' \right]^{-1}\end{aligned}$$

$$x \left\{ \text{Rvec} \left(\tilde{\Omega}_{2,1}^{1/2'} \int dW \tilde{M}' \left(\int \tilde{M} \tilde{M}' \right)^{-1} \Omega_{11}^{-1/2} + \tilde{\Omega}_{12}' \tilde{\Omega}_{11}^{-1} \Gamma \right) \right\}$$

where $\tilde{W}(\lambda)$ is a $r \times 1$ standard Brownian Motion independent of $M(\lambda)$.

This result extends the results of Elliott (1998) to hypothesis testing in the near cointegrating model when stationary variables are added to the system. The Wald test has a nonstandard distribution due to the term $\tilde{\Omega}_{12}' \tilde{\Omega}_{11}^{-1} \Gamma$. When this term is equal to zero, then conditional on $\int \tilde{M} \tilde{M}'$ this distribution is equivalent to a χ_q^2 where the number of degrees of freedom q is equal to the number of restrictions (row rank of R).

The added stationary covariates z_t will be orthogonalizing covariates when $\tilde{\Omega}_{12}' = 0$, i.e. the condition to be satisfied is that $\Omega_{12} - \Omega_{13} \Omega_{33}^{-1} \Omega_{23}' = 0$. This is the zero frequency analog of the same condition for the predictive regression—indeed because of the serial correlation structure of the model in Section 3 the zero frequency variance covariance matrix and the contemporaneous variance covariance matrix coincide, so the restriction is seen to be exactly the same as that for the previous section.

When the control variables z_t are orthogonalizing covariates then this result allows standard t tests to be conducted on β , even when we do not know the value for ρ .

5. Finding and interpreting orthogonalizing regressors

In each of the above cases, predictive regressions and the near cointegrating regressions, the condition for the orthogonalizing covariates is essentially the same. We require that the orthogonalizing covariates z_t be mean reverting and that they explain away the correlation between ε_{1t} and ε_{2t} . In the cointegrating case this becomes the zero frequency correlation.

As with finding instruments in instrumental variables regression, candidate orthogonalizing covariates would generally arise from theoretical understanding of the problem from economic theory and the requirements of the candidate variables from econometric theory. Candidate variables would be those we think are contemporaneously correlated with the shocks to both the predictor variable and to the variable to be predicted. Consider the problem of predicting stock market returns with the log of the dividend price ratio. Here $y_{1t} = \log(\text{Dividends}) - \log(\text{Price})$ and y_{2t} is returns, which also depends on the price of the stock. With prices appearing in both, it is no surprise that $\delta < 0$ and large, indeed δ is often estimated in the range of -0.75 to -0.95 . In Table 1 p. 382 of Stambaugh (1999) they are less than -0.89 . Candidate orthogonalizing regressors would be variables that are correlated with both the dividend price ratio and returns. Consider news variables, for example good macroeconomic news. This would likely have the effect of raising stock prices and hence returns, at the same time lowering the dividend price ratio as prices of the stock rise. Hence such a variable would go some way in explaining this negative correlation.

Thus candidate variables for z_t would come from theoretical considerations for any application. There is no requirement of understanding causal direction, since it is only the correlation that matters for the variable to be potentially useful. Finding such candidates should not be particularly difficult, in the sense that in economic problems contemporaneous correlations between variables are more the norm than orthogonality. Having the candidates actually explain all of the correlation so that $\tilde{\delta} = 0$ is likely more difficult (as is finding instruments that we are sure are uncorrelated with error terms) however the results of this paper show that this is a direction to consider in applied work.

Unlike the instrumental variables problem, in this problem we are able to test the assumptions of the model. In terms of mean reversion of z_t , there exist many tests for this null hypothesis (see Stock, 1994, for a review). Consistent estimators for Ω or Σ can be computed using standard methods and hence can easily estimate both δ and $\tilde{\delta}$ for any application. Consistency follows directly from the asymptotic negligibility of the regression estimates in estimating variance covariance matrices. Hence for any application one could avoid situations where adding z_t to the regression makes the problem worse by not employing the methods unless $|\tilde{\delta}|$ is estimated to be significantly less than the estimate for $|\delta|$.

The effect of introducing z_t to the prediction regression need not be to completely orthogonalize the regression. However as Theorem 2 above showed the effective distribution does change. For situations where $|\tilde{\delta}|$ is much smaller than $|\delta|$, size distortions though not removed will be made much smaller. In such cases it may be a large enough difference at the calculated test statistic that the rejection becomes clear (since for no ρ would the critical value be so high). Hence rather than solving the inference problem, it may instead be that the introduction of z_t merely increases power. This too is interesting, especially if it causes the t statistic to change in such a way that it is now large enough that it is beyond the critical values for all possible γ .

The use of z_t in the prediction regression or the cointegrating model improves power through a number of channels. Here we compare two possibilities—that ρ is known and z_t are not employed versus the more realistic situation that ρ is unknown but also we have and use orthogonalizing covariates z_t .

Theorem 5. When there exist z_t that satisfy the conditions for being orthogonalizing covariates then

- (a) (Predictive Case) Under the conditions of Theorem 2 then the OLS estimator for β employing z_t as a predictor is asymptotically at least as efficient as the asymptotically efficient estimators for β when ρ is known and the information in z_t is not employed.
- (b) (Cointegrating Case) Under the conditions of Theorem 3 then the estimator (13) is asymptotically at least as efficient as the asymptotically efficient estimators for β when ρ is known and the information in z_t is not employed.

In the case of the predictive regression, it is of course not possible that any of the methods discussed in the review in Section 2 that both control size for all ρ and allow uncertainty over this parameter to have better power than a test based on a t -test in the infeasible SUR regression given the efficiency properties of the latter regression for this model. Theorem 5 shows that when orthogonalizing covariates are available, then power using orthogonalizing covariates must be at least as good as the efficient infeasible SUR method that ignores the additional covariates. From this it follows directly that none of the methods presented in Section 2 can have better power against any alternative for β than the method suggested in this paper when z_t is such that $\tilde{\delta} = 0$. A similar point holds in the cointegrating case.

Finally, a comment on interpretation of the regression coefficient. Typically in regression we think of a coefficient such as β as the effect of y_{1t} holding constant other variables in the regression. Thus it might seem that adding the 'orthogonalizing' regressor z_t to the regression fundamentally alters the effect we are trying to estimate. This in turn implies the correct interpretation of the difference in results from choosing to include or exclude the covariate z_t may be that the effect that is being estimated is different in each of the regressions. In the prediction regression, supposing that we knew α_2 , this means that when z_t is excluded we interpret β as the effect of y_{2t-1} on y_{1t} and when z_t is included this coefficient is the

Table 1
Size properties for one sided test.

Method	Regression (8)			Regression (9)			Campbell Yogo		
	$\delta = -0.95$	-0.75	-0.5	-0.95	-0.75	-0.5	-0.95	-0.75	-0.5
$\rho = 1$	0.416	0.289	0.180	0.051	0.051	0.051	0.066	0.067	0.065
$\rho = 0.9$	0.139	0.116	0.091	0.053	0.053	0.053	0.060	0.060	0.058
$\rho = 0.8$	0.105	0.092	0.079	0.054	0.054	0.054	0.061	0.059	0.055

Notes: Size is 5% for a one sided upper tail test.

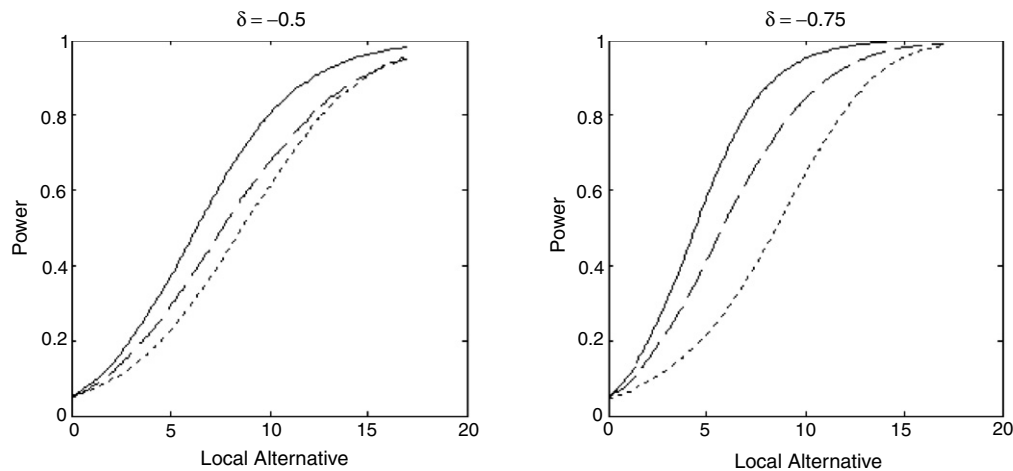


Fig. 2. Local power. The solid line is the method of this paper, long dashes indicate power when ρ is known, short dashes indicate power for the Campbell and Yogo (2006) Bonferroni procedure.

effect of y_{2t-1} on $(y_{2t} - \alpha_2 z_t)$, a different question. For the cointegration case the same would be true for the effect of y_{1t} on y_{2t} . Such an interpretation is however is not correct.

To see this, consider the cointegrating regression results. The interpretation of β is that it is the coefficient that allows the persistent component in $y_{2t} - \beta y_{1t}$ to be annihilated. Adding a stationary variable – one with serial correlation properties that are such that roots are far from the unit circle – cannot possibly change this interpretation of the coefficient since it cannot affect this trending behavior in y_{1t} and y_{2t} . A similar argument holds for the prediction regression. Consider for the prediction regression what it must mean for β to be nonzero. Under this alternative hypothesis, by virtue of the serial correlation properties of y_{2t} , there must be a highly serially correlated component in y_{1t} . This can be true even if standard tests for unit roots or stationarity do not detect this component.⁴ Provided that the orthogonalizing regressor z_t does not have such a low frequency component, it will not affect the relationship between the low frequency behavior of y_{2t} and any equivalent low frequency behavior or lack thereof in y_{2t} .

6. Numerical evaluation of the results

We turn to providing a numerical evaluation of the results. All results will be presented for the predictive model since results are qualitatively the same for the cointegrating model however in the cointegrating model magnitudes can be larger (see Elliott, 1998), hence results presented here are conservative in this respect. For the Monte Carlo results we employ the model used in Campbell and Yogo (2006), which is the first two equations of (7) with $k =$

1, $d_t = 1$ (i.e. constants only, this is the leading applied case for this regression), $\xi = 0$ and $A(L) = 1$. Values for ρ and δ follow their paper as well. For the augmented model we add the third equation of (7) and set $B(L) = 1$, $|\Sigma_{13}| = |\Sigma_{23}|$ so that $\tilde{\delta} = 0$.

We report size results for testing the hypothesis that $\beta = 0$ versus the alternative that $\beta > 0$ with a 5% test in Table 1. The first three columns of results give rejection rates for using the least squares method (8) with use of the normal distribution to construct the critical values for the test. The next three columns of results show size results for the method augmented by including z_t in (9). We also include in the last three columns size for the Campbell and Yogo (2006) Bonferroni method. In each case we used 100 observations and 10 000 Monte Carlo replications.

It is clear from Table 1 that the use of asymptotic normal critical values – the original Stambaugh problem – results in very large size distortions. This was also shown to be true asymptotically in Fig. 1 in Section 2. Having orthogonalizing covariates allows for size to be controlled. This is true for all values of ρ and δ . We can compare this to size results for the method of Campbell and Yogo (2006), where the methods are slightly oversized and the extent of the problem depends on both ρ and δ , where the degree of oversize was increasing in $|\delta|$ and larger for ρ closer to one.

In Fig. 2 we examine power for testing the null of $\beta = 0$ versus the alternative that $\beta > 0$ using the method of this paper, the infeasible approach of knowing ρ , and the Campbell and Yogo (2006) method.⁵ The model is the same as that for Table 1 however we use 250 observations to ensure that all methods control size. The alternative is set to $\beta = b/T$ for b given on the x-axis of the curves. The Bonferroni method of Campbell and Yogo (2006) does not require knowledge of ρ , and by construction gives up power for not knowing this nuisance parameter. Hence we expect that from

⁴ Power of stationarity tests against alternatives where the integrated or near integrated component is small is low. Similarly, unit root tests over-reject when there is a small unit root component mixed in with a large degree of stationary noise.

⁵ Notice that the comparison with the Campbell and Yogo (2006) method involves different levels of information, so is not a fair comparison from this point of view.

the results of Theorem 5 that the augmented regression (9) will result in a test that is more powerful than the infeasible known ρ method from ((4)) which in turn will be more powerful than the Campbell and Yogo (2006) method. The first panel examines power for the model where $\gamma = 10$ and $\delta = -0.5$, the second panel repeats this exercise for $\delta = -0.75$. Similar unreported results hold for $\gamma = 0$ and $\gamma = 20$. In each of these figures the results we expect from Theorem 5 are borne out, with power of the method suggested in this paper being greater than that of the infeasible SUR method which in turn is an upper bound on the ρ unknown bivariate methods. Different choices of Σ for any δ can make power for the orthogonalizing regressor method and the infeasible SUR method closer or further apart.

7. Conclusion

This paper analyzes how control variables that are not persistent can be employed in linear regressions with persistent regressors to reduce or eliminate size distortions of hypothesis tests for the coefficients on these variables. In particular interest is focussed on inference on the coefficients on the persistent covariates. It is shown that in the context of both predictive regressions (where a persistent regressor is employed to forecast a variable that may or may not be persistent) and for near cointegrating regressions (where we suspect that the trends in each of the variables is annihilated by a linear combination but we are unsure if the trend is an exact unit root process) that the addition of control variables to the regression may result in standard asymptotic analysis for the coefficients on the trending regressors even when we do not know the precise specification of the trend itself. Further, when such orthogonalizing control variables are available, inference is improved above and beyond that available through ignoring the control variables but knowing the exact nature of the trend. Finding candidate orthogonalizing covariates – discussed in Section 5 – follows from economic theory and considering what variables are likely to be correlated with the shocks drive the variables in the regression.

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Appendix

For all results $\int$ refers to $\int_0^1$ and $\Rightarrow$ refers to weak convergence.

Lemma 1 (Asymptotic Approximations). *For the model in (7) when Condition 1 holds or (11) when Condition 2 holds then*

- (a) $T^{-2} \sum y_{1t-1} y'_{1t-1} \Rightarrow \Omega_{11}^{1/2} \int M(\lambda) M(\lambda)' d\lambda \Omega_{11}^{1/2}$
- (b) $T^{-1} \sum y_{1t-1} \varepsilon'_t \Rightarrow \Omega_{11}^{1/2} \int M(\lambda) dW(\lambda) \Sigma^{1/2}$
- (c) $T^{-3/2} \sum y_{1t-1} \tilde{d}'_t \Rightarrow \Omega_{11}^{1/2} \int M(\lambda) s(\lambda)' d\lambda$
- (d) $T^{-3/2} \sum y_{1t-1} X'_t \rightarrow^p 0$
- (e) $T^{-1} \sum \tilde{d}_t X_t \rightarrow^p 0$
- (f) $T^{-1/2} \sum X_t \varepsilon'_t$ is $O_p(1)$

where X_t are mean zero terms such that $H(L)X_t$ is stationary with $H(L)$ having a finite lag length.

Proof. The results in (a) and (c) follow directly from condition 1 (a) and (b) which yield $T^{-1/2} y_{1T\lambda} \Rightarrow A(1)^{-1} \Sigma_{11}^{1/2} M(\lambda)$ and

the continuous mapping theorem. For (b) the result follows from Phillips (1987).

The result in (d) is the result behind the familiar block diagonality of the denominator matrix in regression between the integrated and non-integrated pieces. To show the result it is sufficient to show it for an arbitrary element of the matrix, take the (j, r) th element where j may equal r . We have

$$T^{-3/2} \left| \sum_{t=2}^T y_{j1t-1} X_{rt} \right| = T^{-3/2} \left| \sum_{t=2}^T X_{rt} \sum_{s=2}^t \rho_j^{t-s-1} u_s \right| \leq T^{-1/2} \max \{ \rho_j, \rho_j^{T-1} \} \left| T^{-1} \sum_{t=1}^T \sum_{s=1}^t X_{rt} u_s \right|.$$

Since $A_{11}(L)$ and $B(L)$ have finite lags then $T^{-1} \sum_{t=1}^T \sum_{s=1}^t X_{rt} u_s$ converges to $\text{Var}(T^{-1/2} \sum_{s=1}^t X_{rt} u_s)$. The result follows then from showing that $T^{-1/2} \max \{ \rho_j, \rho_j^{T-1} \} \rightarrow 0$. For $\rho_j \leq 1$ then $\rho_j > \rho_j^{T-1}$ and $T^{-1/2} \rho_j \rightarrow 0$. For $\rho_j = 1 + T^{-1} \gamma_j > 0$ by rearrangement $T^{-1/2} \rho_j^{T-1} = T^{-1/2} - \gamma_j T^{-3/2} \sum_{s=0}^{T-1} \gamma_j^s \rightarrow 0$ since $T^{-1} \sum_{s=0}^{T-1} \gamma_j^s \rightarrow \int_0^1 e^{\gamma_j \lambda} d\lambda < \infty$.

For (e) first consider any univariate mean zero random variable x_t with finite variance $E[x_t^2]$. Since $\max_{t=1, \dots, T} \tilde{d}_{it}$ is finite by construction for all elements of the vector $\tilde{d}_t$ then $E[\tilde{d}_t x_t]^2$ is finite and $T^{-1} \sum d_t x_t \rightarrow^p 0$ by Chebyshev. We will show that all elements of X_t satisfy the conditions on x_t plus possible a smaller order term. The first $rx1$ elements of X_t are $u_{2t-1} = \iota_2' A(L)^{-1} \varepsilon_{t-1}$ where $\iota_2 = (0'_{kxr}, I_r, 0'_{pxr})'$ which are mean zero since $E[\varepsilon_t] = 0$. Since all roots of $A(L)$ are outside the unit circle this also has finite variance and so the result holds directly. For $\Delta(y_{1t-1} - \varphi_1 d_{t-1})$ note that this is $(1 - \rho_L)(y_{1t-1} - \varphi_1' d_{t-1}) + (1 - \rho)(y_{1t-2} - \varphi_1' d_{t-2})$ which in turn is $\varepsilon_{1t-1} + T^{-1/2}(T(1 - \rho)T^{-1/2}(y_{1t-2} - \varphi_1' d_{t-2}))$. The second piece is of a smaller magnitude than the first. Further ε_{1t-1} has finite variance by Condition 1(a). For $\Delta(y_{2t-1} - \varphi_2 d_{t-1})$ note that this is equal to $\beta \Delta(y_{1t-1} - \varphi_1 d_{t-1}) + u_{2t-1}$ and hence has finite variance by the immediately preceding two results. For $z_t - \varphi_3 d_t$ the result holds by Condition 1(a) directly. For greater lags the results hold for the reasons just given. Hence all elements of $T^{-1} \sum \tilde{d}_t X_t$ converge to zero as stated.

As just shown in the proof for (e) each element of X_t has bounded variance. Since ε_t is a MDS with four moments then via any MDS central limit theorem this term has an asymptotic normal distribution and is thus $O_p(1)$. $\square$

Proof of Theorem 1. Rewrite (8) so that stochastic terms have their deterministic components removed, (throughout define $y_{1t}^d = y_{1t} - \varphi_1 d_t$, $y_{2t}^d = y_{2t} - \varphi_2 d_t$ and $z_t^d = z_t - \varphi_3 d_t$) i.e.

$$\begin{aligned} y_{2t} &= \tilde{\varphi}_2' \tilde{d}_t + \beta' y_{1t-1} + \varepsilon_{2t} \\ &= \tilde{\varphi}_2' d_t + \beta' y_{1t-1}^d + \varepsilon_{2t} \\ &= \theta' Q_t + \varepsilon_{2t} \end{aligned}$$

here $Q_t = [y_{1t-1}^d, \tilde{d}_t']'$ and $\theta = (\beta', \tilde{\varphi}_2')'$.

Define Υ_T to be a diagonal scaling matrix $\Upsilon_T = \text{diag}[T^{-1} I_k, \Upsilon_{dT}]$ where Υ_{dT} is defined in Condition 1. The least squares estimator for θ , denoted $\hat{\theta}$, is such that

$$\begin{aligned} \Upsilon_T(\hat{\theta} - \theta) &= \left(\Upsilon_T^{-1} \sum_{t=2}^T Q_t Q_t' \Upsilon_T^{-1} \right)^{-1} \left(\Upsilon_T^{-1} \sum_{t=2}^T Q_t \varepsilon_{2t} \right) \\ &= \left(\Upsilon_T^{-1} \sum_{t=2}^T Q_t Q_t' \Upsilon_T^{-1} \right)^{-1} \left(\Upsilon_T^{-1} \sum_{t=2}^T Q_t \varepsilon_t' \iota_2 \right) \end{aligned}$$

where $\iota_2 = (0'_k, 1)'$ where 0_j denotes a $j \times 1$ vector of zeros.

For the denominator

$$\begin{aligned} \gamma_T^{-1} \sum_{t=2}^T Q_t Q_t' \gamma_T^{-1} &= \begin{pmatrix} T^{-2} \sum y_{1t-1}^d y_{1t-1}^{d'} & T^{-3/2} \sum y_{1t-1}^d \tilde{d}_t' \\ T^{-1} \sum \tilde{d}_t \tilde{d}_t' & \end{pmatrix} \\ &\Rightarrow \begin{pmatrix} \Omega_{11}^{1/2} \int MM' \Omega_{11}^{1/2'} & \Omega_{11}^{1/2} \int Ms' \\ \int sM \Omega_{11}^{1/2'} & \int ss' \end{pmatrix} \end{aligned}$$

and for the numerator terms, the term is

$$\gamma_T^{-1} \sum_{t=2}^T Q_t \varepsilon_t' l_2 = \begin{pmatrix} T^{-1} \sum y_{1t-1}^d \varepsilon_t' l_2 \\ T^{-1/2} \sum \tilde{d}_t \varepsilon_t' l_2 \end{pmatrix}.$$

Using the results of Lemma 1 (noting that each of these pieces is $O_p(1)$) and the continuous mapping theorem and some rearrangement we have

$$T(\hat{\beta} - \beta) \Rightarrow \Omega_{11}^{-1/2'} \left(\int \bar{M} \bar{M}' \right)^{-1} \int \bar{M} dW' \Sigma^{1/2'} l_2 \quad (15)$$

The limit distribution for the numerator is a weighted average of $\int \bar{M} dW_1$ which has a nonstandard distribution and $\int \bar{M} dW_2$ which is asymptotically mixed normal. The weights follow from the elements of $l_2 \Sigma^{1/2}$ and are $\Sigma_{12}' \Sigma_{11}^{-1/2'}$ and $\Sigma_{2,1}^{1/2}$ respectively.

The Wald test testing the null hypothesis that $\beta = \beta_0$ against $\beta \neq \beta_0$ is given by

$$\begin{aligned} \text{Wald} &= T^2 \hat{\Sigma}_{22}^{-1} (\hat{\beta} - \beta_0)' \left(S_1 \left(\gamma_T^{-1} \sum Q_t Q_t' \gamma_T^{-1} \right)^{-1} S_1' \right)^{-1} (\hat{\beta} - \beta_0) \\ &= \hat{\Sigma}_{22}^{-1} [T(\hat{\beta} - \beta_0)]' \left(S_1 \left(\gamma_T^{-1} \sum Q_t Q_t' \gamma_T^{-1} \right)^{-1} S_1' \right)^{-1} [T(\hat{\beta} - \beta_0)] \end{aligned}$$

where $S_1 = (I_k, 0_{k \times \dim(\tilde{d}_t)})'$ and $\hat{\Sigma}_{22} = T^{-1} \sum_{t=2}^T (y_{2t} - \hat{\theta}' Q_t)^2$, i.e. the sum of squared residuals of the regression (8) divided by the sample size. Consistency of the OLS estimates gives

$$\hat{\Sigma}_{22} = T^{-1} \sum_{t=2}^T \varepsilon_{2t}^2 + o_p(1)$$

which yields $T^{-1} \sum_{t=2}^T \varepsilon_{2t}^2 \rightarrow^p \Sigma_{22}$.

Hence, from (15) and results for $(\gamma_T^{-1} \sum Q_t Q_t' \gamma_T^{-1})^{-1}$ given above, it follows from the continuous mapping theorem and some algebra that

$$\begin{aligned} \text{Wald} &\Rightarrow \Sigma_{22}^{-1} l_2' \Sigma^{1/2} \left(\int \bar{M} dW \right)' \left(\int \bar{M} \bar{M}' \right)^{-1} \left(\int \bar{M} dW' \right) \Sigma^{1/2'} l_2 \\ &= \left(\int \bar{M} dW_1' \Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1/2} + \Sigma_{2,1}^{1/2} \Sigma_{22}^{-1/2} \int \bar{M} dW_2 \right)' \\ &\quad \times \left(\int \bar{M} \bar{M}' \right)^{-1} \left(\int \bar{M} dW_1' \Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1/2} + \Sigma_{2,1}^{1/2} \Sigma_{22}^{-1/2} \int \bar{M} dW_2 \right) \\ &= \left(\int \bar{M} dW_1 \delta + (1 - \delta' \delta)^{1/2} \int \bar{M} dW_2 \right)' \left(\int \bar{M} \bar{M}' \right)^{-1} \\ &\quad \times \left(\int \bar{M} dW_1 \delta + (1 - \delta' \delta)^{1/2} \int \bar{M} dW_2 \right) \end{aligned}$$

where

$$\begin{aligned} \Sigma_{2,1} &= (\Sigma_{22} - \Sigma_{12}' \Sigma_{11}^{-1} \Sigma_{12}) = \Sigma_{22} (1 - \Sigma_{22}^{-1/2} \Sigma_{12}' \Sigma_{11}^{-1} \Sigma_{12} \Sigma_{22}^{1/2}) \\ &= \Sigma_{22} (1 - \delta' \delta)^{1/2} \end{aligned}$$

so the result is as stated in the text. $\square$

Proof of Theorem 2. Rewrite (9) so that stochastic terms have their deterministic components removed, i.e.

$$\begin{aligned} y_{2t} &= \tilde{\varphi}_2' \tilde{d}_t + \beta' y_{1t-1}^d + \alpha_2' Z_t^d + \tilde{\varepsilon}_{2t} \\ &= \theta' Q_t + \tilde{\varepsilon}_{2t} \end{aligned}$$

where $Q_t = [y_{1t-1}^d, \tilde{d}_t', Z_t^{d'}]'$ and $\theta = (\beta', \tilde{\varphi}_2', \alpha_2')'$.

Define γ_T to be a diagonal scaling matrix $\gamma_T = \text{diag}[T^{-1} I_k, \gamma_{dT}, T^{-1/2} I_{p(q+1)}]$ where γ_{dT} is defined in Condition 1. The least squares estimator for θ , denoted $\hat{\theta}$, is such that

$$\begin{aligned} \gamma_T(\hat{\theta} - \theta) &= \left(\gamma_T^{-1} \sum_{t=2}^T Q_t Q_t' \gamma_T^{-1} \right)^{-1} \left(\gamma_T^{-1} \sum_{t=2}^T Q_t \tilde{\varepsilon}_{2t} \right) \\ &= \left(\gamma_T^{-1} \sum_{t=2}^T Q_t Q_t' \gamma_T^{-1} \right)^{-1} \left(\gamma_T^{-1} \sum_{t=2}^T Q_t \varepsilon_t' l_2 \right). \end{aligned}$$

Consider the denominator term

$$\begin{aligned} \gamma_T^{-1} \sum_{t=2}^T Q_t Q_t' \gamma_T^{-1} &= \begin{pmatrix} T^{-2} \sum y_{2t-1} y_{2t-1}' & T^{-3/2} \sum y_{2t-1} \tilde{d}_t' & T^{-3/2} \sum y_{2t-1} Z_t' \\ \cdot & T^{-1} \sum \tilde{d}_t \tilde{d}_t' & T^{-1} \sum \tilde{d}_t Z_t' \\ \cdot & \cdot & T^{-1} \sum Z_t Z_t' \end{pmatrix} \\ &\Rightarrow \begin{pmatrix} \Omega_{22}^{1/2} \int MM' \Omega_{22}^{1/2'} & \Omega_{22}^{1/2} \int Ms' & 0 \\ \int sM \Omega_{22}^{1/2'} & \int ss' & 0 \\ 0 & 0 & \Sigma_Z \end{pmatrix} \end{aligned}$$

where $\Sigma_Z = E[Z_t Z_t']$.

For the numerator terms, the term is

$$\gamma_T^{-1} \sum_{t=2}^T Q_t \varepsilon_t' G' l_2 = \begin{pmatrix} T^{-1} \sum y_{2t-1} \varepsilon_t' G' l_2 \\ T^{-1/2} \sum \tilde{d}_t \varepsilon_t' G' l_2 \\ T^{-1/2} \sum Z_t \varepsilon_t' G' l_2 \end{pmatrix}.$$

Using the results of Lemma 1 (noting that each of these pieces is $O_p(1)$) and the continuous mapping theorem and some rearrangement we have

$$T(\hat{\beta} - \beta) \Rightarrow \Omega_{11}^{-1/2'} \left(\int \bar{M} \bar{M}' \right)^{-1} \int \bar{M} dW' \Sigma^{1/2'} G' l_2.$$

The weights on dW are given by $\Sigma^{1/2'} G' l_2$, which are

$$l_2' G \Sigma^{1/2} = \begin{pmatrix} \tilde{\Sigma}_{12}' \tilde{\Sigma}_{11}^{-1/2'} \\ \tilde{\Sigma}_{2,1}^{1/2} \\ 0 \end{pmatrix}$$

by judicious choice of $\Sigma^{1/2}$ (i.e. the Choleski decomposition of the variance covariance matrix with the ordering $(\varepsilon_{3t}, \varepsilon_{1t}, \varepsilon_{2t})$) where $\tilde{\Sigma}_{2,1} = \tilde{\Sigma}_{22} - \tilde{\Sigma}_{12}' \tilde{\Sigma}_{11}^{-1} \tilde{\Sigma}_{12}$ and $\tilde{\Sigma}$ as defined in (10).

The numerator of the limit distribution has two components, i.e. we can write

$$\int \bar{M} dW' \Sigma^{1/2'} G' l_2 = \int \bar{M} dW_1' \tilde{\Sigma}_{11}^{-1/2'} \tilde{\Sigma}_{12} + \tilde{\Sigma}_{2,1} \int \bar{M} d\tilde{W}_2$$

where the first term has a nonstandard distribution and the second is mixed normal. Hence we have as before

$$\begin{aligned} T(\hat{\beta} - \beta) &\Rightarrow A(1)^{-1} \Sigma_{11}^{-1/2'} \left(\int \bar{M} \bar{M}' \right)^{-1} \\ &\quad \times \left(\int \bar{M} dW_1' \Sigma_{11}^{-1/2'} (\Sigma_{12} - \Sigma_{13} \Sigma_{33}^{-1} \Sigma_{23}') \right. \\ &\quad \left. + \tilde{\Sigma}_{22}^{1/2} (1 - \delta' \delta)^{1/2} \int \bar{M} d\tilde{W}_2 \right) \end{aligned}$$

conditional on $(\int \bar{M}\bar{M}')^{-1}$.

The Wald test testing hypotheses β for β_0 is

$$\begin{aligned} \text{Wald} &= \hat{\Sigma}_{22}^{-1}(\hat{R}\hat{\theta} - r)' \left(R \left(\sum Q_t Q_t' \right)^{-1} R' \right)^{-1} (\hat{R}\hat{\theta} - r) \\ &= \hat{\Sigma}_{22}^{-1} [T(\hat{\beta} - \beta_0)]' \left(R \left(\gamma_T^{-1} \sum Q_t Q_t' \gamma_T^{-1} \right)^{-1} R' \right)^{-1} \\ &\quad \times [T(\hat{\beta} - \beta_0)] \end{aligned}$$

where $R = (R_\beta, 0, 0)$ and $r = \beta_0$ where the partitions of R are conformable with θ and $\hat{\Sigma}_{22} = T^{-1} \sum_{t=2}^T (y_{2t} - \hat{\theta}' Q_t)^2$, i.e. the sum of squared residuals of the regression (9) divided by the sample size. By analogous results to those in Theorem 1 $\hat{\Sigma}_{22} = T^{-1} \sum_{t=2}^T \tilde{\varepsilon}_{2t}^2 + o_p(1)$ and application of Chebyshev using assumption (fourth moments) yields $T^{-1} \sum_{t=2}^T \tilde{\varepsilon}_{2t}^2 \rightarrow^p \Sigma_{22} - \Sigma_{23}' \Sigma_{33}^{-1} \Sigma_{23} = \tilde{\Sigma}_{22}$.

Hence, using the asymptotic results for $T(\hat{\beta} - \beta_0)$ and $(\gamma_T^{-1} \sum Q_t Q_t' \gamma_T^{-1})^{-1}$ given above, it follows from the continuous mapping theorem and some algebra that

$$\begin{aligned} \text{Wald} &\Rightarrow \tilde{\Sigma}_{22}^{-1} l_2' G \Sigma^{1/2} \left(\int \bar{M} dW \right)' \left(\int \bar{M} \bar{M}' \right)^{-1} \\ &\quad \times \left(\int \bar{M} dW' \right) \Sigma^{1/2'} G' l_2 \\ &= \left(\int \bar{M} dW_1' \tilde{\delta} + (1 - \tilde{\delta}' \tilde{\delta})^{1/2} \int \bar{M} d\tilde{W}_2 \right) \left(\int \bar{M} \bar{M}' \right)^{-1} \\ &\quad \times \left(\int \bar{M} dW_1' \tilde{\delta} + (1 - \tilde{\delta}' \tilde{\delta})^{1/2} \int \bar{M} d\tilde{W}_2 \right) \end{aligned}$$

as stated in the text.

For the distribution to have an asymptotic mixed normal distribution we require the weight on $(\int \bar{M} \bar{M}')^{-1} \int \bar{M} dW_1$ to be zero which occurs only if $\Sigma_{12}' - \Sigma_{23} \Sigma_{33}^{-1} \Sigma_{13}' = 0$. Under the restriction $\Sigma_{12}' - \Sigma_{23} \Sigma_{33}^{-1} \Sigma_{13}' = 0$

$$\begin{aligned} T(\hat{\beta} - \beta) &\Rightarrow \Omega_{11}^{-1/2'} \left(\int \bar{M} \bar{M}' \right)^{-1} \int \bar{M} dW' \Sigma^{1/2'} G' l_2 \\ &\sim {}^a N \left(0, \tilde{\Sigma}_{22} \left(\int \bar{M} \bar{M}' \right)^{-1} \right). \end{aligned}$$

When z_t satisfies the conditions for being an orthogonalizing regressor, i.e. $\Sigma_{12}' - \Sigma_{23} \Sigma_{33}^{-1} \Sigma_{13}' = 0$ holds, then conditional on $\int \bar{M} \bar{M}'$ this is a χ_k^2 random variable. $\square$

Proof of Theorem 3. Define the matrix

$$P = \begin{pmatrix} I_k & 0 & 0 \\ \beta & I_r & 0 \\ 0 & 0 & I_p \end{pmatrix}.$$

The model (11) can be written

$$P \tilde{A}(L) P^{-1} P \begin{pmatrix} (1 - \rho L) y_{1t}^d \\ y_{2t}^d - \beta y_{1t}^d \\ z_t^d \end{pmatrix} = P \varepsilon_t = \varepsilon_t^*.$$

On rearrangement this yields

$$\begin{aligned} P \tilde{A}(L) P^{-1} \begin{pmatrix} \Delta(y_{1t}^d) + (I_k - \rho)(y_{1t-1}^d) \\ \Delta(y_{2t}^d) + (y_{2t-1}^d - \beta y_{1t-1}^d) + \beta(I - \rho)(y_{1t-1}^d) \\ z_t^d \end{pmatrix} \\ = P \varepsilon_t = \varepsilon_t^*. \end{aligned}$$

Defining $\tilde{A}(1) = \sum \tilde{A}_i$ where $\tilde{A}_i$ are the matrices of coefficients in $\tilde{A}(L)$ and the summation is over all the lags (including the zero lag) and rearranging we can write this as

$$\begin{pmatrix} \Delta y_{1t} \\ \Delta y_{2t} \\ z_t \end{pmatrix} = \underline{\Psi}_1 y_{1t-1}^d + \underline{\Psi}_2 (y_{2t-1}^d - \beta y_{1t-1}^d) \quad (16)$$

$$+ \Phi^* d_t + \Pi(L) \begin{pmatrix} \Delta y_{1t-1}^d \\ \Delta y_{2t-1}^d \\ z_t^d \end{pmatrix} + \varepsilon_t^* \quad (17)$$

where $\underline{\Psi}_1 = P \tilde{A}(1) l_1 (\rho - I_k)$ and $\underline{\Psi}_2 = -P \tilde{A}(1) l_2$ and we define $l_1 = (I_k, 0'_{kxr}, 0'_{pxr})'$ and $l_2 = (0'_{kxr}, I_r, 0'_{pxr})'$ are selector matrices. This regression is equivalent to the regression (12) apart from rearranging the deterministic components so each of the regressors has this term subtracted out and we have rotated the term

$$\begin{aligned} \Psi \begin{pmatrix} y_{1t-1} \\ y_{2t-1} \end{pmatrix} &= \Psi_1 y_{1t-1} + \Psi_2 y_{2t-1} \\ &= (\Psi_1 + \Psi_2 \beta) y_{1t-1} + \Psi_2 (y_{2t-1} - \beta y_{1t-1}) \\ &= \underline{\Psi}_1 y_{1t-1} + \underline{\Psi}_2 (y_{2t-1} - \beta y_{1t-1}) \end{aligned}$$

where Ψ_1 is $(k+r+p) \times k$, Ψ_2 is $(k+r+p) \times r$ and $\underline{\Psi}_1 = \Psi_1 + \Psi_2 \beta$.

The regression is in the Sims et al. (1990) format that allows limit results to be obtained. Define the matrix

$$X_t = \begin{pmatrix} (y_{2t-1} - \varphi_2 d_{t-1}) - \beta (y_{1t-1} - \varphi_1 d_{t-1}) \\ \Delta(y_{1t-1} - \varphi_1 d_{t-1}) \\ \Delta(y_{2t-1} - \varphi_2 d_{t-1}) \\ z_{t-1} - \varphi_3 d_{t-1} \\ \vdots \end{pmatrix}$$

where the ...are the remaining lags of each of the detrended changes in Δy_{1t} , Δy_{2t} and z_t . Define $Q_t = [y_{1t-1}^d, d_t', X_t']$ and collect the coefficients of the regression into θ , the regression can be written

$$\begin{pmatrix} \Delta y_{1t} \\ \Delta y_{2t} \\ z_t \end{pmatrix} = \theta' Q_t + \varepsilon_t^*.$$

Define the diagonal scaling matrix γ_T partitioned conformably with the rows of Q_t and equal to $[T^{-1} I_k, \gamma_{Tr}, T^{-1/2} I_g]$ where g is the number of rows in X_t . Then

$$\begin{aligned} \gamma_T (\hat{\theta} - \theta) &= \left(\gamma_T^{-1} \sum Q_t Q_t' \gamma_T^{-1} \right)^{-1} \left(\gamma_T^{-1} \sum Q_t \varepsilon_t^* \right) \\ &= \left(\gamma_T^{-1} \sum Q_t Q_t' \gamma_T^{-1} \right)^{-1} \left(\gamma_T^{-1} \sum Q_t \varepsilon_t^* P' \right) \end{aligned}$$

where the summations are over available observations once (the finite number of) lags have been accounted for. Consider the denominator term

$$\begin{aligned} \gamma_T^{-1} \sum Q_t Q_t' \gamma_T^{-1} &= \begin{pmatrix} T^{-2} \sum y_{1t-1} y_{1t-1}' & T^{-3/2} \sum y_{1t-1} \tilde{d}_t' & T^{-3/2} \sum y_{1t-1} X_t' \\ \cdot & T^{-1} \sum \tilde{d}_t \tilde{d}_t' & T^{-1} \sum \tilde{d}_t X_t' \\ \cdot & \cdot & T^{-1} \sum X_t X_t' \end{pmatrix} \\ &\Rightarrow \begin{pmatrix} \Omega_{11}^{1/2} \int M(\lambda) M(\lambda)' d\lambda \Omega_{11}^{1/2'} & \Omega_{11}^{1/2} \int M s(\lambda)' d\lambda & 0 \\ \int s(\lambda) M(\lambda) d\lambda \Omega_{11}^{1/2'} & \int s(\lambda) s(\lambda)' d\lambda & 0 \\ 0 & 0 & K \end{pmatrix} \end{aligned}$$

where K is a (matrix) constant. The result for the (1, 1) and (1, 2) elements follows from Lemma 1(a) and (c) respectively. The result in the (2, 2) element of the matrix is Condition 1(e). The result in

(3, 3) follows from the stationarity of X_t and in the (2, 3) element from Lemma 1(e).

For the numerator

$$\gamma_T^{-1} \sum_{t=2}^T Q_t \varepsilon_t' P' s_2 = \begin{pmatrix} T^{-1} \sum y_{1t-1} \varepsilon_t' P' \\ T^{-1/2} \sum \tilde{d}_t \varepsilon_t' P' \\ T^{-1/2} \sum X_t \varepsilon_t' P' \end{pmatrix}.$$

For the first of these terms $T^{-1} \sum y_{1t-1} \varepsilon_t' P' \Rightarrow \Omega_{11}^{-1/2} \int M dW' \Sigma^{1/2'} P'$ from Lemma 1(b). Similarly for the second term $T^{-1/2} \sum \tilde{d}_t \varepsilon_t' P' \Rightarrow \int s(\lambda) dW'(\lambda) d\lambda \Sigma^{1/2'} P'$. Finally, we have from Lemma 1(f) that $T^{-1/2} \sum X_t \varepsilon_t' P'$ is $O_p(1)$. By the asymptotic block diagonality of the denominator and the boundedness of the third term of the numerator, along with algebra as above we have

$$T(\hat{\Psi}'_1 - \underline{\Psi}'_1) \Rightarrow \Omega_{11}^{-1/2'} \left(\int \bar{M} \bar{M}' \right)^{-1} \int \bar{M} dW' \Sigma^{1/2'} P'.$$

The convergence of $T^{-1} \sum X_t X_t'$ to the constant K and the boundedness of $T^{-1/2} \sum X_t \varepsilon_t' P'$ means that the coefficients on X_t are all consistently estimated, hence we have that $\hat{\Psi}_2 \rightarrow^p \Psi_2$. Since all coefficients converge to their true values then $\hat{\Sigma}^* = T^{-1} \sum \hat{\varepsilon}_t^* \hat{\varepsilon}_t^{*'} \rightarrow^p \Sigma^* = P \Sigma P'$ where $\hat{\varepsilon}_t^*$ are the least squares residuals from the regression (12).

Define the $(k+r) \times (k+r)$ matrix $\tilde{\Omega}$ where

$$\tilde{\Omega} = \begin{pmatrix} \Omega_{11} - \Omega_{13} \Omega_{33}^{-1} \Omega_{13}' & \Omega_{12} - \Omega_{13} \Omega_{33}^{-1} \Omega_{23}' \\ \Omega_{12}' - \Omega_{23} \Omega_{33}^{-1} \Omega_{13}' & \Omega_{22} - \Omega_{23} \Omega_{33}^{-1} \Omega_{23}' \end{pmatrix}.$$

Notice that this is the inverse of the upper $(k+r) \times (k+r)$ block of Ω^{-1} .

Recall $\Psi_2 = P \tilde{A}(1) \iota_2$ then

$$\begin{aligned} (\hat{\Psi}_2' \hat{\Sigma}^{*-1} \hat{\Psi}_2) &\rightarrow^p (\iota_2' \tilde{A}(1)' P' (P \Sigma P')^{-1} P \tilde{A}(1) \iota_2) \\ &= (\iota_2' \tilde{A}(1)' \Sigma^{-1} \tilde{A}(1) \iota_2) \\ &= (\iota_2' \Omega^{-1} \iota_2) \\ &= \tilde{\Omega}_{2,1}^{-1} \end{aligned}$$

where $\tilde{\Omega}_{2,1} = \tilde{\Omega}_{22} - \tilde{\Omega}_{12}' \tilde{\Omega}_{11}^{-1} \tilde{\Omega}_{12}$.

From the definition of the suggested estimator (13) then

$$\begin{aligned} T(\hat{\beta} - \beta) &= -(\hat{\Psi}_2' \hat{\Sigma}^{*-1} \hat{\Psi}_2)^{-1} (\hat{\Psi}_2' \hat{\Sigma}^{*-1} T(\hat{\Psi}_1 + \hat{\Psi}_2 \beta)) \\ &= -(\hat{\Psi}_2' \hat{\Sigma}^{*-1} \hat{\Psi}_2)^{-1} (\hat{\Psi}_2' \hat{\Sigma}^{*-1} T(\hat{\Psi}'_1 - \underline{\Psi}'_1)) \\ &\quad - (\hat{\Psi}_2' \hat{\Sigma}^{*-1} \hat{\Psi}_2)^{-1} (\hat{\Psi}_2' \hat{\Sigma}^{*-1} T \underline{\Psi}'_1) \\ &\Rightarrow \tilde{\Omega}_{2,1} \iota_2' \Omega^{-1/2'} \int dW \bar{M}' \left(\int \bar{M} \bar{M}' \right)^{-1} \\ &\quad \times \Omega_{11}^{-1/2} + \tilde{\Omega}_{12}' \tilde{\Omega}_{11}^{-1} \Gamma \end{aligned}$$

where $(\Psi_2' \Sigma^{*-1} \Psi_2)^{-1} (\Psi_2' \Sigma^{*-1} T \underline{\Psi}'_1) = -\tilde{\Omega}_{2,1} \iota_2' \Omega^{-1} \iota_1 \Gamma = -\tilde{\Omega}_{12}' \tilde{\Omega}_{11}^{-1} \Gamma$.

The stochastic part of this result has a mixed normal distribution. The weights $\iota_2' \Omega^{-1/2'}$ on $\int dW \bar{M}'$ are

$$\Omega^{-1/2} \iota_2 = \begin{pmatrix} 0 \\ \tilde{\Omega}_{2,1}^{-1} \\ -\Omega_{3,21}^{-1/2} (\Omega_{23} - \Omega_{13}' \Omega_{11}^{-1} \Omega_{12}') \tilde{\Omega}_{2,1}^{-1} \end{pmatrix}$$

where $\Omega_{3,21}^{-1} = \Omega_{33}^{-1} (I + [\Omega_{13}', \Omega_{23}'] \tilde{\Omega}^{-1} [\Omega_{13}', \Omega_{23}']' \Omega_{33}^{-1})$ and so there is a zero weight on $\int dW_1 \bar{M}'$ which is the element with a nonstandard distribution. It follows directly that the variance covariance matrix of this sum of mixed normals is $\iota_2' \Omega^{-1} \iota_2 =$

$\tilde{\Omega}_{2,1}^{-1}$ so $\tilde{\Omega}_{2,1}^{1/2} \iota_2' \Omega^{-1/2'} \int dW \bar{M}' \left(\int \bar{M} \bar{M}' \right)^{-1}$ is free of nuisance parameters. $\square$

Proof of Theorem 4. For Wald tests of hypotheses of the form $\text{Rvec}(\beta) = r$ the test statistic is

$$\begin{aligned} \text{Wald} &= T^2 (\text{Rvec}(\hat{\beta}) - r)' \left[R(\hat{M} \otimes (\hat{\Psi}_2' \hat{\Sigma}^{*-1} \hat{\Psi}_2)^{-1})^{-1} R' \right]^{-1} \\ &\quad \times (\text{Rvec}(\hat{\beta}) - r) \\ &\Rightarrow \left\{ \text{Rvec} \left(\tilde{\Omega}_{2,1} \iota_2' \Omega^{-1/2'} \int dW \bar{M}' \left(\int \bar{M} \bar{M}' \right)^{-1} \right. \right. \\ &\quad \times \left. \left. \Omega_{11}^{-1/2} + \tilde{\Omega}_{12}' \tilde{\Omega}_{11}^{-1} \Gamma \right) \right\}' \\ &\quad \times \left[R \left(\Omega_{11}^{-1/2'} \left(\int \bar{M} \bar{M}' \right)^{-1} \Omega_{11}^{-1/2} \otimes \tilde{\Omega}_{2,1} \right) R' \right]^{-1} \\ &\quad \cdot \left\{ \text{Rvec} \left(\tilde{\Omega}_{2,1} \iota_2' \Omega^{-1/2'} \int dW \bar{M}' \right. \right. \\ &\quad \times \left. \left. \left(\int \bar{M} \bar{M}' \right)^{-1} \Omega_{11}^{-1/2} + \tilde{\Omega}_{12}' \tilde{\Omega}_{11}^{-1} \Gamma \right) \right\} \end{aligned}$$

where $\hat{M}$ is the upper left $k \times k$ block of $\gamma_T^{-1} \sum Q_t Q_t' \gamma_T^{-1}$. This distribution has a center that depends on the nuisance parameters of the model. When z_t are orthogonalizing covariates $\tilde{\Omega}_{12} = 0$ and this collapses to the distribution

$$\begin{aligned} \text{Wald} &\Rightarrow \left\{ \text{Rvec} \left(\tilde{\Omega}_{2,1} \iota_2' \Omega^{-1/2'} \int dW \bar{M}' \left(\int \bar{M} \bar{M}' \right)^{-1} \Omega_{11}^{-1/2} \right) \right\}' \\ &\quad \cdot \left[R(\Omega_{11}^{-1/2'} \left(\int \bar{M} \bar{M}' \right)^{-1} \Omega_{11}^{-1/2} \otimes \tilde{\Omega}_{2,1}) R' \right]^{-1} \\ &\quad \cdot \left\{ \text{Rvec} \left(\tilde{\Omega}_{2,1} \iota_2' \Omega^{-1/2'} \int dW \bar{M}' \left(\int \bar{M} \bar{M}' \right)^{-1} \Omega_{11}^{-1/2} \right) \right\}. \end{aligned}$$

Conditional on $\int \bar{M} \bar{M}'$ this distribution is equivalent to a χ_q^2 where the number of degrees of freedom q is equal to the number of restrictions (row rank of R).

Finally, $\iota_2' \Omega^{-1/2'} W(\lambda)$ is $rx1$ with variance covariance matrix $\iota_2' \Omega^{-1/2'} \Omega^{-1/2} \iota_2 \lambda = \iota_2' \Omega^{-1} \iota_2 \lambda = \tilde{\Omega}_{2,1}^{-1} \lambda$ so is equivalent to $\tilde{\Omega}_{2,1}^{-1/2'} \tilde{W}(\lambda)$ where $\tilde{W}(\lambda)$ is a standard $rx1$ Brownian Motion. This allows the simplified notation

$$\begin{aligned} \text{Wald} &\Rightarrow \left\{ \text{Rvec} \left(\tilde{\Omega}_{2,1}^{1/2'} \int d\tilde{W} \tilde{M}' \left(\int \tilde{M} \tilde{M}' \right)^{-1} \Omega_{11}^{-1/2} \right) \right\}' \\ &\quad \cdot \left[R(\Omega_{11}^{-1/2'} \left(\int \tilde{M} \tilde{M}' \right)^{-1} \Omega_{11}^{-1/2} \otimes \tilde{\Omega}_{2,1}) R' \right]^{-1} \\ &\quad \cdot \left\{ \text{Rvec} \left(\tilde{\Omega}_{2,1}^{1/2'} \int d\tilde{W} \tilde{M}' \left(\int \tilde{M} \tilde{M}' \right)^{-1} \Omega_{11}^{-1/2} \right) \right\}. \quad \square \end{aligned}$$

Proof of Theorem 5. We will prove the result for the cointegrating case, the result for the predictive case is identical replacing Ω with Σ . Asymptotically efficient estimators for β in the absence of z_t are such that $T(\text{vec}(\hat{\beta} - \beta))$ are asymptotically mixed normal with variance covariance matrix $\Omega_{2,1} \otimes \left(\Omega_{11}^{1/2} \int \bar{M} \bar{M}' \Omega_{11}^{1/2'} \right)^{-1}$. It follows directly from Theorem 4 that when ρ is unknown however we do have orthogonalizing covariates then $T(\text{vec}(\hat{\beta} - \beta))$

are asymptotically mixed normal with variance covariance matrix $\tilde{\Omega}_{22} \otimes \left(\Omega_{11}^{1/2} \int \bar{M} \bar{M}' \Omega_{11}^{1/2'} \right)^{-1}$ where we have used the restriction that $\tilde{\Omega}_{12} = 0$ for z_t that satisfy the conditions for being orthogonalizing covariates. Hence the latter estimator is asymptotically more efficient if $\Omega_{2,1} - \tilde{\Omega}_{22}$ is positive definite. We have

$$\begin{aligned} \Omega_{2,1} - \tilde{\Omega}_{22} &= (\Omega_{22} - \Omega_{12}' \Omega_{11}^{-1} \Omega_{12}) - (\Omega_{22} - \Omega_{23} \Omega_{33}^{-1} \Omega_{23}') \\ &= \Omega_{23} \Omega_{33}^{-1} \Omega_{23}' - \Omega_{12}' \Omega_{11}^{-1} \Omega_{12} \\ &= \Omega_{23} \Omega_{33}^{-1} (\Omega_{33} - \Omega_{13}' \Omega_{11}^{-1} \Omega_{13}) \Omega_{33}^{-1} \Omega_{23}' \end{aligned}$$

where the last line follows as for z_t to be an orthogonalizing covariate $\Omega_{12} = \Omega_{13}' \Omega_{33}^{-1} \Omega_{23}'$. Now $\Omega_{33} - \Omega_{13}' \Omega_{11}^{-1} \Omega_{13}$ is the variance covariance matrix of a zero frequency projection of z_t on u_{1t} and is hence positive definite, Ω_{33}^{-1} is positive definite, and so this quadratic is positive semi definite and the result holds. $\square$

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